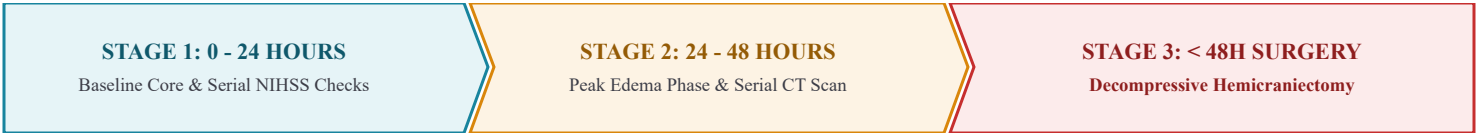


Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.



1. Decompressive Hemicraniectomy Selection Criteria

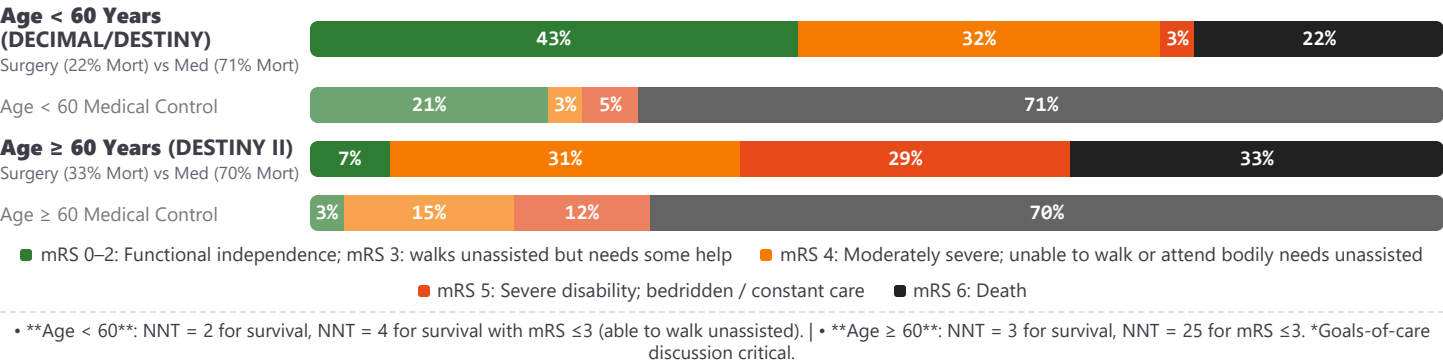
Clinical Deficit Severity:

- NIHSS > 15 (non-dominant hemisphere)
- NIHSS > 20 (dominant hemisphere)
- AND decrease in level of consciousness (NIHSS Item 1a score ≥ 1 / obtunded or stuporous)
- **Timing**:** Surgery performed **within 48 hours** of onset.

Radiographic Markers:

- Infarction of ≥ 50% of the MCA territory (CT/MRI)
- DWI core volume > 82 mL within 6 hours
- DWI core volume > 145 mL within 14 hours
- Midline shift or mass effect on repeat imaging
- **Surgical Spec**:** Bone flap diameter ≥ 12–15 cm with duraplasty.

2. Surgical Outcomes & Evidence (By Age Group)



3. Supportive ICU Care & Medical Management

Positioning	Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.
Fluids	Maintain euvolemia with isotonic fluids. Avoid hypotonic fluids (e.g. D5W, 0.45% NS) that can worsen edema; balanced crystalloids such as LR should follow local neuro-ICU protocol.
Osmotherapy	Consider targeted PRN hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. <i>Prophylactic osmotherapy is not recommended.</i>
Metabolic	Target normothermia (<37.8°C). Target normocapnia (PaCO2 35-45 mmHg); avoid hypoventilation/hypercapnia.
Steroids	Class III (Harmful): Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.

DECIMAL: Vahedi K et al. *Stroke*. 2007;38:2506-2517. | DESTINY: Jüttler E et al. *Stroke*. 2007;38:2518-2525.

HAMLET: Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. | DESTINY II: Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100.

AHA Guidelines: Wijdicks EF et al. *Stroke*. 2014;45:1222-1238.